Example 7.7.2. The category of modules over an algebra A and the category of finite dimensional modules over A are abelian categories.

Problem 7.7.3. Let A be a finitely generated commutative ring. Show that the category of finitely generated A-modules is an abelian category.

Hint: Use the Hilbert basis theorem.

We see from this definition that in an abelian category, $\mathrm{Hom}(X, Y)$ is an abelian group for each X, Y, compositions are group homomorphisms with respect to each argument, there is the zero object, the notions of an injective morphism (monomorphism) and surjective morphism (epimorphism), and every morphism has a kernel, a cokernel, and an image.

Remark 7.7.4. The good thing about Definition 7.7.1 is that it allows us to visualize objects, morphisms, kernels, and cokernels in terms of classical algebra. But the definition also has a big drawback, which is that even if $\mathcal{C}$ is the whole category A-mod, the ring A is not determined by $\mathcal{C}$. In particular, two different rings can have equivalent categories of modules (such rings are called **Morita equivalent**). Actually, it is worse than that: for many important abelian categories there is no natural (or even manageable) ring A at all. This is why people prefer to use the standard definition, which is free from this drawback, even though it is more abstract.

Let k be a field. We say that an abelian category $\mathcal{C}$ is **k-linear** if the groups $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ are equipped with a structure of a vector space over k and composition maps are k-linear in each argument. In particular, the categories in Example 7.7.2 are k-linear.

7.8. Complexes and cohomology

Definition 7.8.1. A sequence of objects $C_i, i \in \mathbb{Z}$, of an abelian category $\mathcal{C}$ and morphisms $d_i : C_i \to C_{i+1}$ is said to be **a complex** if the composition of any two consecutive arrows is zero. The morphisms d_i are called the **differentials**.[1] The **cohomology** of this complex is

[1] A famous example of a complex is the **de Rham complex**, in which C_m is the space of differential m-forms on a Euclidean space or, more generally, a manifold, and d_m is the exterior differential of differential forms. This explains the term "differential".